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United States Patent	12410869
Kind Code	B2
Date of Patent	September 09, 2025
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Two-pin clapper check valve

Abstract

In some implementations, a clapper valve assembly includes a valve body defining a longitudinal flow bore, a hanger disposed in the valve body, a link piece coupled to the hanger, a first pin coupled between the hanger and the link piece, wherein the link piece is configured to rotate, relative to the hanger, about the first pin, a clapper coupled to the link piece, and a second pin coupled between the link piece and the clapper, wherein the clapper is configured to rotate, relative to the link piece, about the second pin.

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Appl. No.:	18/456553
Filed:	August 28, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240117889 A1	Apr. 11, 2024

Related U.S. Application Data

us-provisional-application US 63378544 20221006

Publication Classification

Int. Cl.: F16K15/03 (20060101)

U.S. Cl.:

CPC **F16K15/03** (20130101);

Field of Classification Search

CPC: F16K (15/03); F16K (15/033); F16K (15/034)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This Patent Application claims priority to U.S. Patent Application No. 63/378,544, filed on Oct. 6, 2022. The disclosure of the prior Application is considered part of and is incorporated by reference into this Patent Application.

TECHNICAL FIELD

(1) The present disclosure relates generally to a check valve and, for example, to a clapper valve assembly with a two-pin design.

BACKGROUND

(2) In oil and gas operations (e.g., production of a subterranean wellbore, hydraulic fracturing operations, or other operations), a check valve (e.g., clapper valve) may be used to control the flow of fluids (e.g., production fluids, hydraulic fracturing fluids, or other fluids) by preventing backflow. The clapper may stay open during forward flow and close, as a result of gravitation and/or pressure differential, during backflow. Dynamic loading and/or abrasion caused by fluid flow through the clapper valve (e.g., including the effects of flow dynamics that cause cavitation and cyclic loading on parts) may cause significant wear to internal components of the clapper valve. Moreover, friction and wear caused by linear and/or rotary movement of parts (e.g., pins, pin holes, slots, or other parts) may shorten the life of those parts. Replacement and/or repair of worn parts is inefficient, time-consuming, and costly.

(3) The apparatus of the present disclosure solves one or more of the problems set forth above and/or other problems in the art.

SUMMARY

(4) In some implementations, a clapper valve assembly includes a valve body defining a longitudinal flow bore; a hanger disposed in the valve body and including a slot; a link piece, including a link body; and first and second pin holes defined in the link body; a first pin disposed in both the slot of the hanger and the first pin hole of the link body, wherein the first pin defines a first rotation axis, and wherein the link piece is configured to rotate about the first rotation axis; a clapper, including a clapper body; a third pin hole defined in the clapper body; and a seal; a second pin disposed in both the second pin hole of the link body and the third pin hole of the clapper body, wherein the second pin defines a second rotation axis, and wherein the clapper is configured to rotate about the second rotation axis; and a valve seat disposed in the valve body.

(5) In some implementations a link assembly for coupling a clapper to a hanger of a clapper valve includes a link piece, including a link body; and first and second pin holes defined in the link body; a first pin disposed in the first pin hole of the link body, wherein the first pin defines a first rotation axis, and wherein the link piece is configured to rotate, relative to the hanger, about the first rotation axis; and a second pin disposed in the second pin hole of the link body, wherein the second pin defines a second rotation axis, and wherein the clapper is configured to rotate, independently of the link piece, about the second rotation axis.

(6) In some implementations a clapper valve assembly includes a valve body defining a longitudinal flow bore; a hanger disposed in the valve body; a link piece coupled to the hanger; a first pin coupled between the hanger and the link piece, wherein the link piece is configured to rotate, relative to the hanger, about the first pin; a clapper coupled to the link piece; and a second pin coupled between the link piece and the clapper, wherein the clapper is configured to rotate, relative to the link piece, about the second pin.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a diagram of a perspective section view of an example clapper valve assembly described herein.

(2) FIG. 2 is a diagram of a side section view of the example clapper valve assembly, described herein, in an open position.

(3) FIG. 3 is a diagram of a side section view of the example clapper valve assembly, described herein, in a closed position.

(4) FIG. 4 is a diagram of a side section view of the example clapper valve assembly, described herein, in a hard stop position.

DETAILED DESCRIPTION

(5) This disclosure relates to a clapper valve assembly, which is applicable to any process that controls the flow of fluids (e.g., production fluids, hydraulic fracturing fluids, or other fluids). The term “process” may refer to an operation associated with an industry such as, for example, oil and gas, refining, mining, construction, farming, transportation, or another industry. The clapper valve assembly, described herein, uses a two-pin design with a link piece to reduce friction and wear. In addition, the clapper valve assembly, described herein, may use materials (e.g., tool steel for all components) that reduce wear. As a result, replacement and/or repair of the clapper valve assembly is reduced, which drives efficiency, time savings, and cost savings.

(6) FIG. 1 is a diagram of a perspective section view of an example clapper valve assembly **100** described herein. In some examples, the clapper valve assembly **100** may include less equipment, additional equipment, or alternative equipment compared to the example depicted in FIG. 1.

(7) The clapper valve assembly **100** may include a valve body **102**, a hanger **104** disposed in the valve body **102**, a link piece **106**, a first pin **108**, a clapper **110**, a second pin **112**, a valve seat **114**

disposed in the valve body **102**, and a cap **116**.

(8) The valve body **102** may define a longitudinal flow bore **118**. The valve body **102** may include an inlet **120** and an outlet **122** defined at opposite ends of the longitudinal flow bore **118**. The longitudinal flow bore **118** may be in fluid communication with flow lines (e.g., conduits, piping, tubing, or other flow lines) connected upstream of the inlet **120** and downstream of the outlet **122**. The inlet **120** and outlet **122** may include connections (e.g., flanged, bolted, threaded, or other connections) configured to be coupled to the flow lines.

(9) The valve body **102** may be formed (e.g., cast, machined, or otherwise formed) as a single piece or assembled from multiple pieces. The valve body **102** may be formed from a steel material (e.g., carbon steel, alloy steel, tool steel, or other steel materials). The valve body **102** may include a connection **124** (e.g., threaded, fastened, or other connections), defined in the longitudinal flow bore **118**, to couple the valve seat **114** to the valve body **102**. The connection **124** may include a recess. The valve body **102** may include a transverse bore **126**, defined perpendicular to the longitudinal flow bore **118**, to receive the cap **116** in the valve body **102**. The transverse bore **126** may include a connection to couple the cap **116** to the valve body **102**.

(10) The hanger **104** may include a hanger body **128** configured to be coupled to the valve body **102**. The hanger body **128** may be annular in shape having a step that rests on a corresponding shoulder of the transverse bore **126**. The hanger body **128** may include a hinge section disposed along at least a radial portion thereof. The hanger body **128** may include a slot **130** (see FIG. 2, inset) defined in the hinge section of the hanger body **128**. The slot **130** may be oblong in shape (e.g., oval, rectangle, or other oblong shapes) to allow linear movement (e.g., sliding) of the first pin **108** within the slot **130** in a direction parallel to the longitudinal flow bore **118**.

(11) The link piece **106** may be coupled to the hanger **104** via the first pin **108**. The link piece **106** may include a link body **132** and first and second pin holes **134**, **136** defined in the link body **132**. The first and second pin holes **134**, **136** may be disposed parallel to each other and/or perpendicular to the longitudinal flow bore **118**. The second pin hole **136** may have a smaller inner diameter than the first pin hole **134**.

(12) The link body **132** may include a first end **138** defining a rounded edge corresponding to the first pin hole **134**. The link body **132** has a first width, measured parallel to the first pin hole **134**, and a first depth d_1 (see FIG. 3) at the first end **138**. The link body **132** may include a second end **140** corresponding to the second pin hole **136**. The link body **132** has a second width, measured parallel to the second pin hole **136**, and a second depth d_2 (see FIG. 3) at the second end **140**. The second width may be less than the first width at least because the link body **132** may be narrower towards the second end **140** in order to fit inside a corresponding notch in the clapper **110**.

Likewise, the second depth d_2 may be less than the first depth d_1 (see FIG. 3).

(13) The link body **132** may include a forward-facing surface **142** connecting the first and second ends **138**, **140**. The forward-facing surface **142** may be configured to face towards the outlet **122** of the longitudinal flow bore **118** when the clapper **110** is in a closed position (see FIG. 3). The link body **132** may include a backward-facing surface **144** connecting the first and second ends **138**, **140**. The backward-facing surface **144** may be configured to face towards the inlet **120** of the longitudinal flow bore **118** when the clapper **110** is in the closed position (see FIG. 3). A length of the backward-facing surface **144** defined between the rounded edge and the second end **140** may be less than a length of the forward-facing surface **142**. The first and second widths of the link body **132** may be defined, parallel to the first and second pin holes **134**, **136**, in relation to either the forward-facing surface **142**, the backward-facing surface **144**, or both.

(14) The first pin **108** may be disposed in both the slot **130** of the hanger **104** and the first pin hole **134** of the link body **132**. The first pin **108** may define a first rotation axis **146**. The first rotation axis **146** may be perpendicular to the longitudinal flow bore **118**. The link piece **106** may be configured to rotate, relative to the hanger **104**, about the first rotation axis **146** via the first pin **108**.

- (15) The clapper **110** may include a clapper body **148**, a third pin hole **150** (shown in phantom) defined in the clapper body **148**, and a seal **152** coupled to the clapper body **148**. The third pin hole **150** may be disposed parallel to the first and second pin holes **134**, **136** and/or perpendicular to the longitudinal flow bore **118**. The third pin hole **150** may have an equal inner diameter to the second pin hole **136**.
- (16) The clapper body **148** may be formed (e.g., cast, machined, or otherwise formed) as a single piece or assembled from multiple pieces. As shown in FIG. **1**, the clapper body **148** includes a hinge **154** coupled to the link piece **106**, via the second pin **112**, and an insert **156** coupled to the hinge **154**. During operation, after the seal **152** is compressed, a face **158** of the insert **156** may be configured to make final contact with the valve seat **114** in a hard stop position (see FIG. **4**).
- (17) The seal **152** may be fit (e.g., sandwiched, compressed, or other fits) between the hinge **154** and the insert **156**. The seal **152** may be annular in shape to seal around a circumference of the longitudinal flow bore **118**. The seal may be formed from an elastomeric material (e.g., urethane or other elastomers).
- (18) The second pin **112** may be disposed in both the second pin hole **136** of the link body **132** and the third pin hole **150** of the clapper body **148**. The second pin **112** may define a second rotation axis **160**. The second rotation axis **160** may be perpendicular to the longitudinal flow bore **118**. The clapper **110** may be configured to rotate, relative to the link piece **106**, about the second rotation axis **160** via the second pin **112**. The first and second pins **108**, **112** may be disposed parallel to each other. The second pin **112** may be nearer a centerline of the longitudinal flow bore **118**, compared to the first pin **108**, when the clapper **110** is in the closed position (see FIG. **3**). The second pin **112** may have a smaller outer diameter than the first pin **108**.
- (19) In some examples, all of the internal components of the clapper valve assembly **100** that may experience friction and/or wear caused by one or more of fluid flow induced dynamic loading and/or abrasion or linear and/or rotary movement of parts (e.g., the hanger **104**, link piece **106**, first pin **108**, clapper **110**, and second pin **112**) may be formed from the same material (e.g., tool steel) to reduce wear.
- (20) The valve seat **114** may be coupled to the valve body **102** via the connection **124**. The valve seat **114** may be disposed in the recess of the longitudinal flow bore **118**. The valve seat **114** may include an opening aligned with the longitudinal flow bore **118**. The valve seat **114** may include a face **162** configured to make initial contact with the seal **152** in the closed position (see FIG. **3**). The face **162** may be configured to make final contact with the face **158** of the insert **156** in the hard stop position (see FIG. **4**).
- (21) As indicated above, FIG. **1** is provided as an example. Other examples may differ from what was described in connection with FIG. **1**.
- ### INDUSTRIAL APPLICABILITY
- (22) FIG. **2** is a diagram of a side section view of the example clapper valve assembly **100**, described herein, in an open position. In FIG. **2**, the valve body **102** is hidden for clarity. In the inset of FIG. **2**, the link piece **106** is hidden for clarity. FIG. **3** is a diagram of a side section view of the example clapper valve assembly **100**, described herein, in a closed position. In FIG. **3**, the valve body **102** is hidden for clarity. FIG. **4** is a diagram of a side section view of the example clapper valve assembly **100**, described herein, in a hard stop position. The portion of the clapper valve assembly **100** shown in FIG. **4** corresponds to the inset of FIG. **3**. In some examples, the clapper valve assembly **100** may include less equipment, additional equipment, or alternative equipment compared to the example depicted in FIGS. **2-4**.
- (23) As shown in the inset of FIG. **2**, linear movement of the first pin **108**, within the slot **130**, may occur in a direction parallel to the longitudinal flow bore **118** (e.g., right/left in FIG. **2**). In some examples, a first fit between the first pin **108** and the slot **130** of the hanger **104** may be looser than a bearing fit (e.g., clearance fit or other fits). Linear movement of the first pin **108** may be configured for self-alignment between the clapper **110** and the valve seat **114** to improve sealing.

For example, self-alignment may compensate for geometrical deviations resulting from accumulation of geometrical tolerances during manufacturing.

(24) For the clapper **110** to move from the open position (see FIG. 2) to the closed position (see FIG. 3), the link piece **106** may be configured to rotate, relative to the hanger **104**, about the first rotation axis **146** of the first pin **108**. In addition, the clapper **110** may be configured to rotate, independently of the link piece **106**, about the second rotation axis **160** of the second pin **112**. In some examples, a second fit between the second pin **112** and the third pin hole **150** of the clapper body **148** may be a bearing fit. In some examples, the first fit between the first pin **108** and the slot **130** of the hanger **104** may have greater clearance than the second fit between the second pin **112** and the third pin hole **150** of the clapper body **148**. In some examples, the configuration of the link piece **106**, described herein, may enable tighter clearances between parts, such that loading, abrasion, and/or wear effects, particularly associated with greater clearances between moving parts, may be reduced. For example, the first fit may be a bearing fit, such that both the first and second fits are bearing fits, thereby reducing cumulative clearances between moving parts.

(25) In some examples, independent rotation of the link piece **106** and the clapper **110** may occur simultaneously and/or in sequential order. Independent rotation of the link piece **106** and the clapper **110** may cause linear movement of the clapper **110** between the closed position (see FIG. 3) and the hard stop position (see FIG. 4). In other words, linear movement of the clapper **110** may be configured to occur from initial contact between the seal **152** and the valve seat **114** (closed position) to final contact between a face **158** of the clapper body **148** and the valve seat **114** (hard stop position). In some examples, linear movement of the clapper **110** may be configured to occur during compression of the seal **152**. For example, compression of the seal **152** is shown in FIG. 4 where a dashed line is used to illustrate a portion of the seal **152** in the uncompressed state. Linear movement of the clapper **110** may be, and/or defined, in a direction normal to the valve seat **114** (e.g., parallel to the longitudinal flow bore **118**, or from right to left in FIGS. 3-4). In some examples, linear movement of the clapper **110** may be configured to occur when the clapper **110** is oriented perpendicular to the longitudinal flow bore **118** (e.g., fully vertical, or up and down in FIGS. 3-4). In some examples, linear movement of the clapper **110** may be configured to occur without sliding of the first pin **108** within the slot **130** (e.g., based only on the configuration of the link piece **106** with two pins, described herein, or in combination with only rotation of the first pin **108** being allowed and linear movement of the first pin **108** being prevented, such as when the slot **130** is replaced with a fourth pin hole).

(26) In some examples, linear movement of the clapper **110** may be configured for self-alignment between the clapper **110** and the valve seat **114** to improve sealing. In some examples, self-alignment caused by linear movement of the clapper **110**, based on rotation about the second rotation axis **160** of the second pin **112**, may replace the self-alignment function associated with linear movement of the first pin **108** within the slot **130** (e.g., based only on the configuration of the link piece **106** with first and second pins **108**, **112**, described herein, whether or not linear movement of the first pin **108** is prevented). This modification, from sliding of the first pin **108** to rotation about the second pin **112**, based on the configuration of the link piece **106**, described herein, may reduce friction and extend part life.

(27) As indicated above, FIGS. 2-4 are provided as an example. Other examples may differ from what was described in connection with FIGS. 2-4.

(28) In oil and gas operations, a clapper valve may be used to control the flow of fluids by preventing backflow. Dynamic loading and/or abrasion caused by fluid flow through the clapper valve (e.g., including the effects of flow dynamics that cause cavitation and cyclic loading on parts) may cause significant wear to internal components of the clapper valve. Moreover, friction and wear caused by linear and/or rotary movement of parts (e.g., pins, pin holes, slots, or other parts) may shorten part life. Replacement and/or repair of worn parts is inefficient, time-consuming, and costly. The clapper valve assembly, and application thereof, described herein, reduces friction and

wear, which drives efficiency, time savings, and cost savings.

(29) The foregoing disclosure provides illustration and description, but is not intended to be exhaustive or to limit the implementations to the precise forms disclosed. Modifications and variations may be made in light of the above disclosure or may be acquired from practice of the implementations. Furthermore, any of the implementations described herein may be combined unless the foregoing disclosure expressly provides a reason that one or more implementations cannot be combined. Even though particular combinations of features are recited in the claims and/or disclosed in the specification, these combinations are not intended to limit the disclosure of various implementations. Although each dependent claim listed below may directly depend on only one claim, the disclosure of various implementations includes each dependent claim in combination with every other claim in the claim set.

(30) As used herein, “a,” “an,” and a “set” are intended to include one or more items, and may be used interchangeably with “one or more.” Further, as used herein, the article “the” is intended to include one or more items referenced in connection with the article “the” and may be used interchangeably with “the one or more.” Further, the phrase “based on” is intended to mean “based, at least in part, on” unless explicitly stated otherwise. Also, as used herein, the term “or” is intended to be inclusive when used in a series and may be used interchangeably with “and/or,” unless explicitly stated otherwise (e.g., if used in combination with “either” or “only one of”). Further, spatially relative terms, such as “below,” “lower,” “above,” “upper,” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the apparatus, device, and/or element in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

Claims

1. A clapper valve assembly, comprising: a valve body defining a longitudinal flow bore; a hanger disposed in the valve body and including a slot; a link piece, including: a link body; and first and second pin holes defined in the link body; a first pin disposed in both the slot of the hanger and the first pin hole of the link body, wherein the first pin defines a first rotation axis, and wherein the link piece is configured to rotate about the first rotation axis; a clapper, including: a clapper body; a third pin hole defined in the clapper body; and a seal; a second pin disposed in both the second pin hole of the link body and the third pin hole of the clapper body, wherein the second pin defines a second rotation axis, wherein the clapper is configured to rotate about the second rotation axis, and wherein a fit between the first pin and the slot of the hanger has greater clearance than a fit between the second pin and the third pin hole of the clapper body; and a valve seat disposed in the valve body.
2. The clapper valve assembly of claim 1, wherein the link piece and the clapper are configured to rotate independently of each other.
3. The clapper valve assembly of claim 2, wherein the independent rotation of the link piece and the clapper causes linear movement of the clapper.
4. The clapper valve assembly of claim 3, wherein the linear movement of the clapper is normal to the valve seat.
5. The clapper valve assembly of claim 3, wherein the linear movement is configured to occur from initial contact between the seal and the valve seat to final contact between a face of the clapper body and the valve seat.
6. The clapper valve assembly of claim 1, wherein the first and second pins are disposed parallel to each other, and wherein the second pin is nearer a centerline of the longitudinal flow bore,

compared to the first pin, when the clapper is in a closed position.

7. The clapper valve assembly of claim 1, wherein the first and second rotation axes are perpendicular to the longitudinal flow bore.

8. The clapper valve assembly of claim 1, wherein the link body includes: a first end defining a rounded edge corresponding to the first pin hole, wherein the link body has a first width and a first depth at the first end; and a second end corresponding to the second pin hole, wherein the link body has a second width and a second depth at the second end, wherein the second width is less than the first width, and wherein the second depth is less than the first depth.

9. The clapper valve assembly of claim 8, wherein the link body further includes: a forward-facing surface connecting the first and second ends, wherein the forward-facing surface is configured to face towards an outlet of the longitudinal flow bore when the clapper is in a closed position; and a backward-facing surface connecting the first and second ends, wherein the backward-facing surface is configured to face towards an inlet of the longitudinal flow bore when the clapper is in the closed position, and wherein a length of the backward-facing surface defined between the rounded edge and the second end is less than a length of the forward-facing surface.

10. A link assembly for coupling a clapper to a hanger of a clapper valve, comprising: a link piece, including: a link body; and first and second pin holes defined in the link body, wherein the link body includes: a first end corresponding to the first pin hole, wherein the link body has a first width and a first depth at the first end; and a second end corresponding to the second pin hole, wherein the link body has a second width and a second depth at the second end, wherein the second width is less than the first width, and wherein the second depth is less than the first depth; a first pin disposed in the first pin hole of the link body, wherein the first pin defines a first rotation axis, and wherein the link piece is configured to rotate, relative to the hanger, about the first rotation axis; and a second pin disposed in the second pin hole of the link body, wherein the second pin defines a second rotation axis, and wherein the clapper is configured to rotate, independently of the link piece, about the second rotation axis.

11. The link assembly of claim 10, wherein the link piece and the clapper are configured to rotate independently of each other.

12. The link assembly of claim 11, wherein the independent rotation of the link piece and the clapper causes linear movement of the clapper.

13. The link assembly of claim 10, wherein the link body further includes: a forward-facing surface connecting the first and second ends, wherein the forward-facing surface is configured to face towards an outlet of the clapper valve when the clapper is in a closed position; and a backward-facing surface connecting the first and second ends, wherein the backward-facing surface is configured to face towards an inlet of the clapper valve when the clapper is in the closed position, and wherein a length of the backward-facing surface defined between a rounded edge, defined at the first end and corresponding to the first pin hole, and the second end is less than a length of the forward-facing surface.

14. A clapper valve assembly, comprising: a valve body defining a longitudinal flow bore; a hanger disposed in the valve body and including a slot; a link piece coupled to the hanger; a first pin coupled between the hanger and the link piece, wherein the link piece is configured to rotate, relative to the hanger, about the first pin; a clapper coupled to the link piece; and a second pin coupled between the link piece and the clapper, wherein the clapper is configured to rotate, relative to the link piece, about the second pin, and wherein a fit between the first pin and the slot of the hanger has greater clearance than a fit between the second pin and a pin hole of the clapper.

15. The clapper valve assembly of claim 14, wherein the link piece and the clapper are configured to rotate independently of each other.

16. The clapper valve assembly of claim 15, wherein the independent rotation of the link piece and the clapper causes linear movement of the clapper.

17. The clapper valve assembly of claim 14, wherein the first and second pins are disposed parallel

to each other, and wherein the second pin is nearer a centerline of the longitudinal flow bore, compared to the first pin, when the clapper is in a closed position.

18. The clapper valve assembly of claim 14, wherein the clapper includes: a clapper body, wherein the pin hole is defined in the clapper body, and wherein the pin hole is configured to receive the second pin; and a seal coupled to the clapper body.
